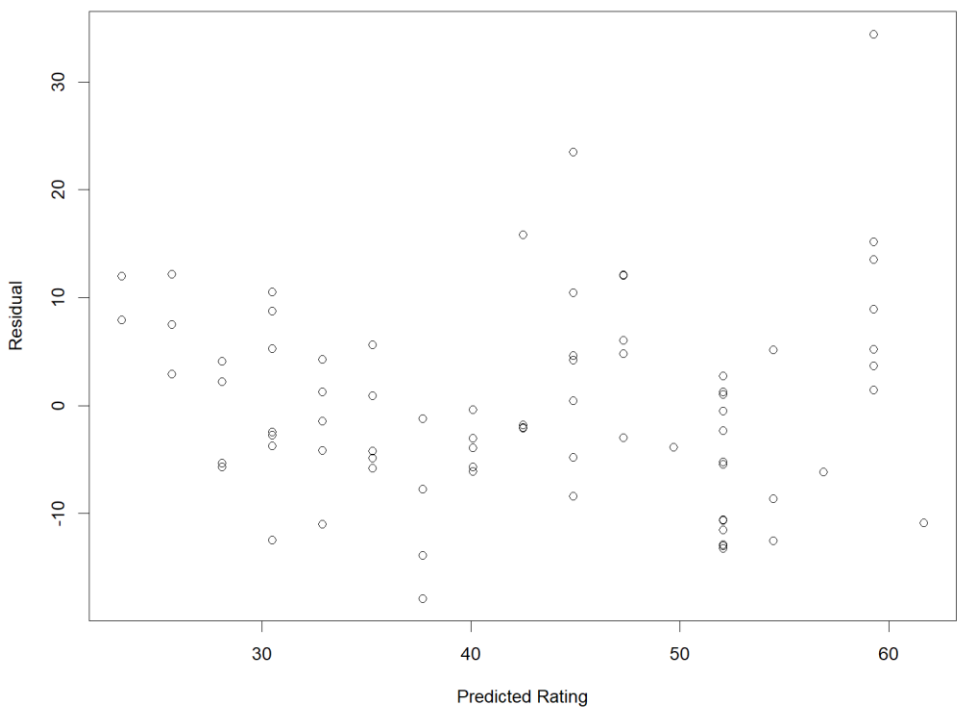
Best Simple Linear Regression Model:
To choose the best simple linear regression model(I was looking in particular for quantitative variables for the predictor variable) I looked at the scatterplots I created for each potential qualitative predictor in the last project along with its correlation coefficient, r. After that I fitted the top 3 models, which I thought showed the strongest linear relationship in R studio. After that, I picked the one with the highest R-squared value, which was the amount of sugars vs cereal rating with a R-squared value of 0.5715, r value of -0.75967466.

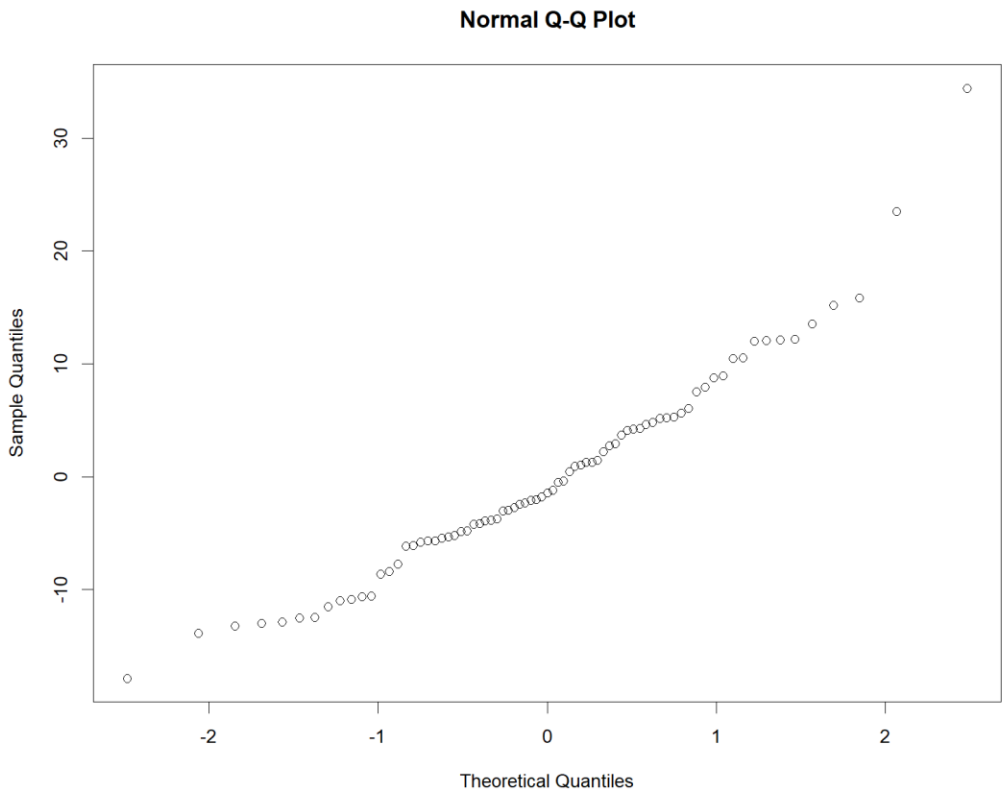
Final Model: y-hat= $59.2844-2.4008x_{sugar}$

Model Assumptions: iid E ~N(0, σ^2), iid= independent and identically distributed

For the “iid” part of the assumption of the errors, it is unclear since we don’t know if the data was collected via random sample.



The residual plot doesn’t seem to show extreme patterns although there may be 1 or 3 outliers, but we can still assume constant variance for the residuals. However, after using cooks' distance to identify influential outliers, none of them were influential.



The QQ plot seems fairly linear, so we can assume normality of the errors.

Slope Interpretation: The estimated mean change in the customer rating decreases by 2.4008 for every one gram of increase in sugar levels of the cereals in grocery stores.

Multiple Linear Regression (Model Selection Method=Stepwise) :

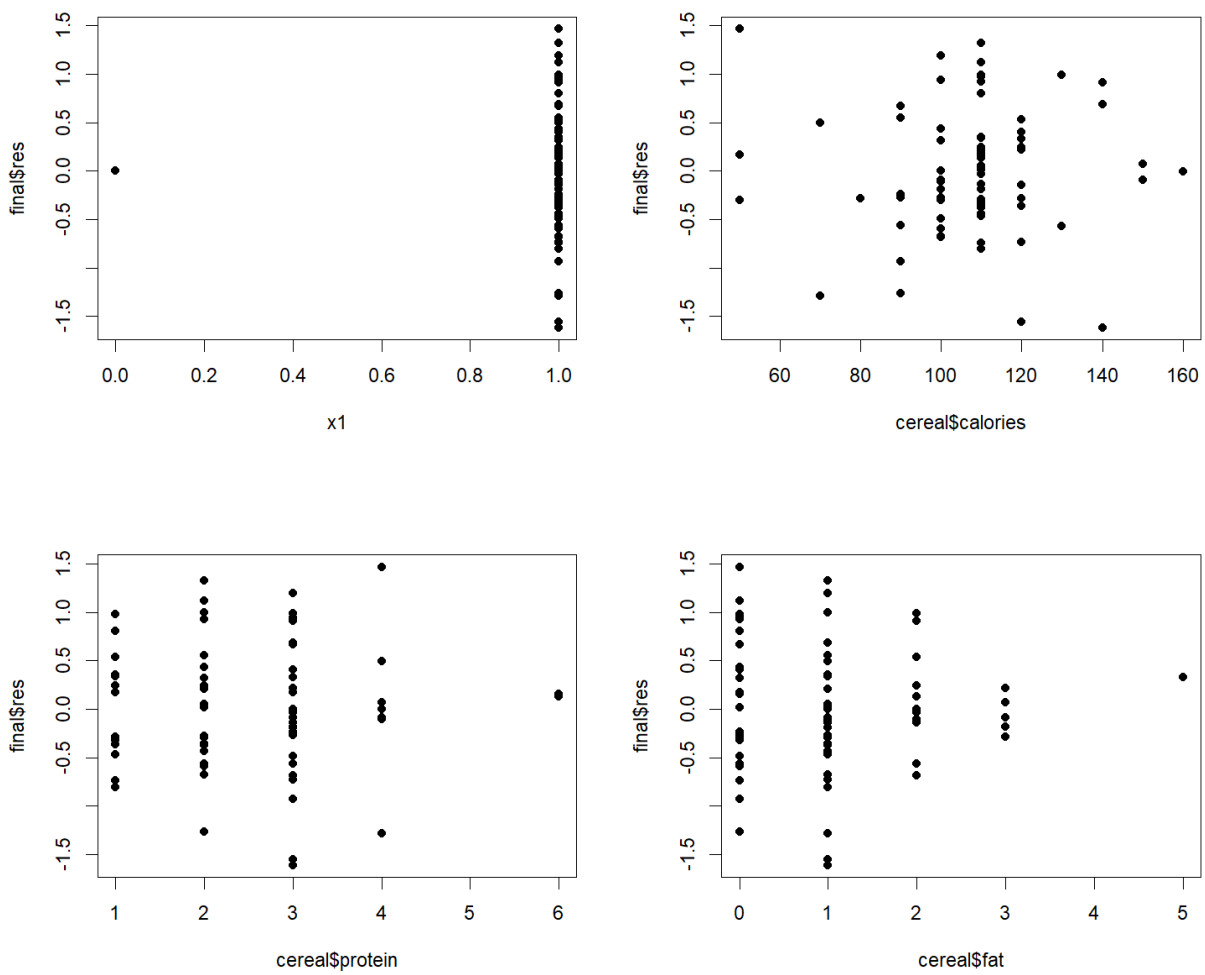
x_1 ={ 1 if Hot, 0 if Cold
x_2 = Calories
x_3 = Protein
x_4 = Fat
x_5 = Sodium
x_6 = Fiber
x_7 = Carbohydrates (carbo)
x_8 = Sugars
x_9 = Potassium (potass)
x_{10} = { 1 if 20% Vitamin, 0 if not
x_{11} = {1 if 100% Vitamin ,0 if not
x_{12} = {1 if on bottom shelf, 0 if not
x_{13} = {1 if on middle shelf, 0 if not
x_{14} = Weight

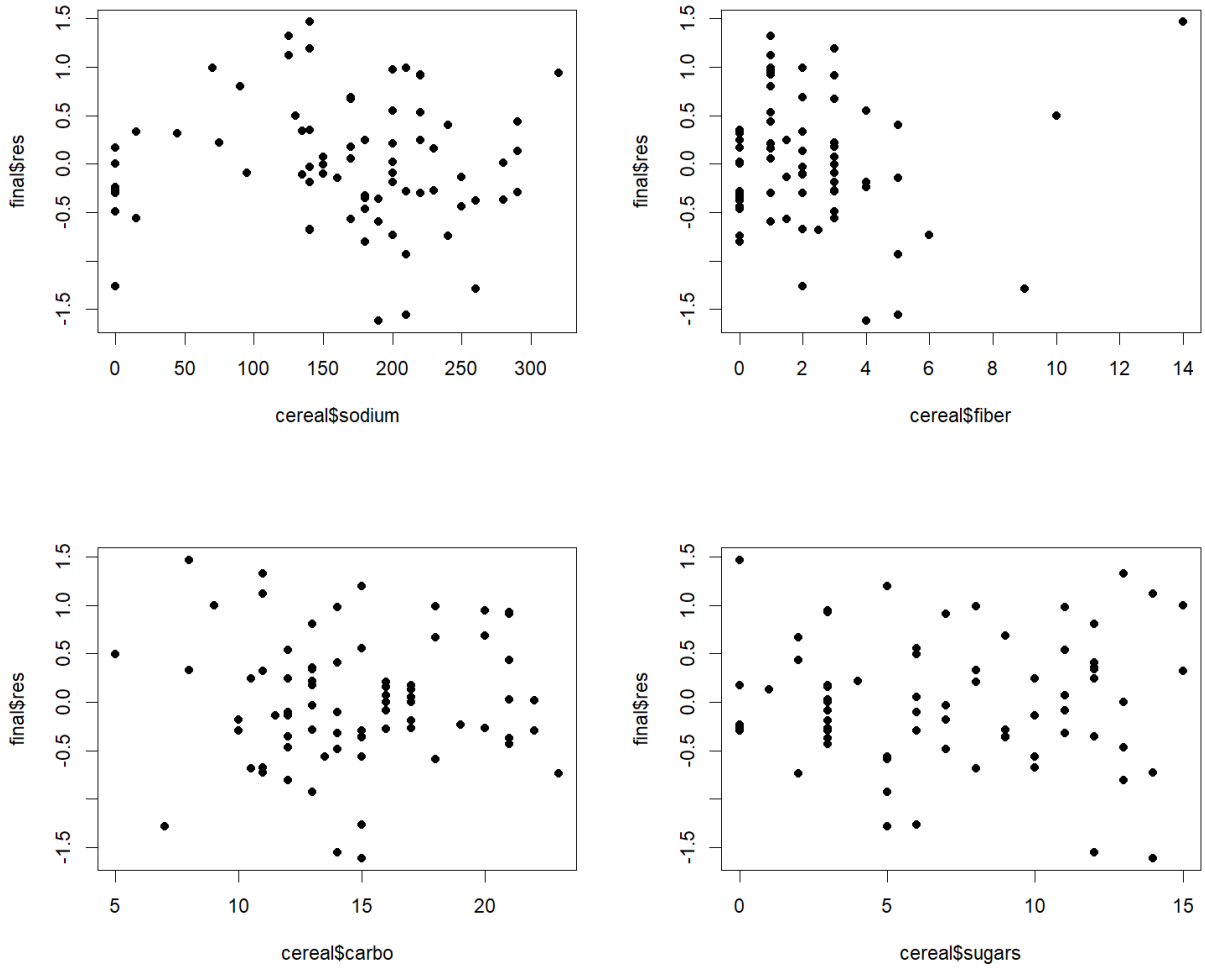
x_{15} = Cups
x_{16}={ 1 if popular brand, 0 if not

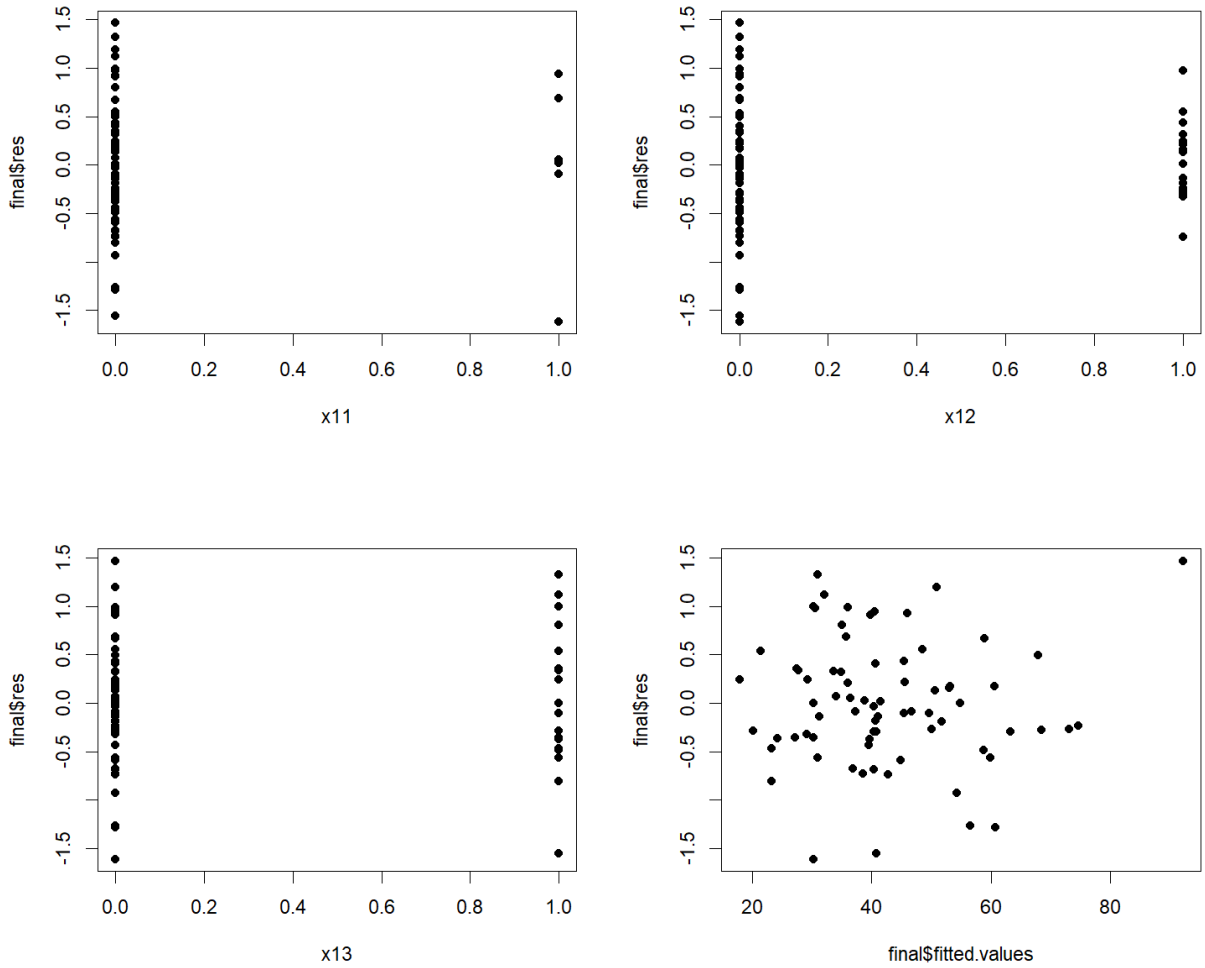
Warning: “attempting model selection on an essentially perfect fit is nonsense”. I got this warning when I was trying to use stepwise model selection to select a good model, suggesting overfitting. This might be due to the fact that I have 16 predictor variables for only 74 individual data. To reduce the number of variables I looked at the correlations between the quantitative variables weight and calories and fiber and potassium and fiber have high correlation, 0.70 and 0.91 respectively, so I will be dropping weight and potassium from the full model.

Chosen Model: y-hat=$52.77371+2.88906x_1-0.19381x_2+2.88755x_3-2.16931x_4 - 0.05739x_5+2.53812x_6+0.90184x_7-0.99990x_8-3.59314x_{11}+0.53874x_{12}+0.64805x_{13}$

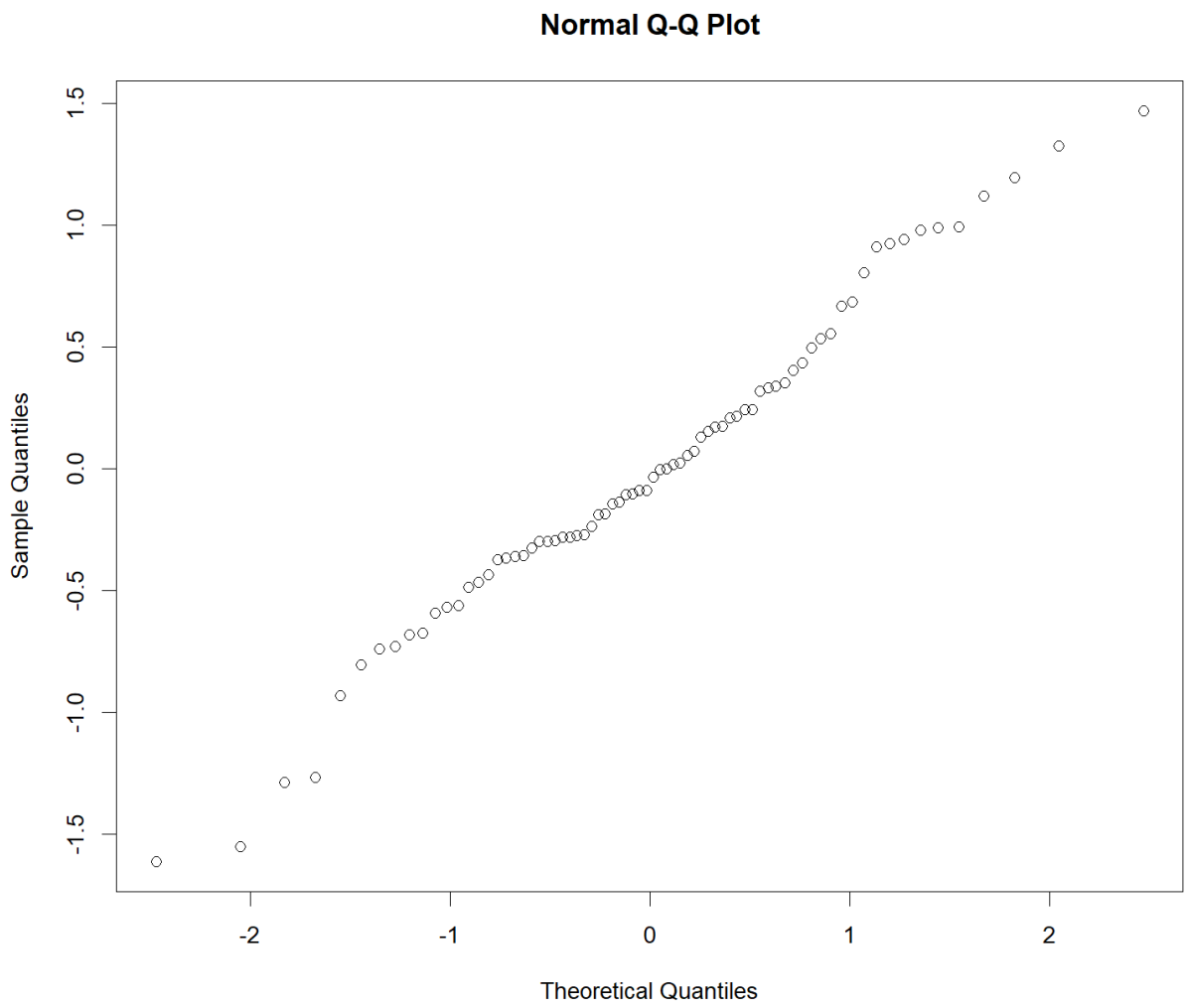
Assumptions: iid $E \sim N(0,\sigma^2)$




final$res
cereal$sodium
cereal$fiber
cereal$carbo
cereal$sugars



The residuals plot for the fitted values versus the residuals of the final multiple regression model doesn’t seem to have a distinct pattern, so we can assume constant variance of the errors.



The normality assumption of the errors can be passed since the QQ plot is fairly linear.

The most significant predictor was fiber. The estimated mean change in customer ratings of cereal increases by 2.53812 for every 1 gram of increase in fiber for the cereals in grocery stores, while the type of cereal, calories, protein, fat, sodium, fiber, carbohydrates, sugars, if it contains 100% vitamin or not, and shelf placements(bottom, middle, top) stays constant.

We are 95% confident that the mean cereal rating increases by 2.43702243 to 2.63920948 for every 1 gram increase in fiber for cereals in grocery stores, while holding type of cereal, calories, protein, fat, sodium, fiber, carbohydrates, sugars, if it contains 100% vitamin or not, and shelf placements(bottom, middle, top) constant.

VIF: The variables sugar, carbohydrates, and calories all have VIF values larger than 10, 14.77, 13.45, and 16.811 respectively, indicating multicollinearity may be causing problems in the estimation of the model. This can be confirmed by the correlation matrix of all the numerical

variables. The variable calories has moderate to strong correlation with fat, sugars, and weight variables. Also, carbohydrates(carbo) has a moderate to weak relationships with sodium, fiber, sugars, potassium, and cups. Then, sugars has moderate to weak relationships with calories, protein, fat, carbohydrates, and weight

	calories	protein	fat	sodium	fiber	carbo	sugars	potass	weight	cups
calories	1.00	0.03	0.51	0.30	-0.30	0.27	0.57	-0.07	0.70	0.09
protein	0.03	1.00	0.20	0.01	0.51	-0.04	-0.29	0.58	0.23	-0.24
fat	0.51	0.20	1.00	0.00	0.01	-0.28	0.29	0.20	0.22	-0.16
sodium	0.30	0.01	0.00	1.00	-0.07	0.33	0.04	-0.04	0.31	0.12
fiber	-0.30	0.51	0.01	-0.07	1.00	-0.38	-0.15	0.91	0.25	-0.51
carbo	0.27	-0.04	-0.28	0.33	-0.38	1.00	-0.45	-0.37	0.14	0.36
sugars	0.57	-0.29	0.29	0.04	-0.15	-0.45	1.00	0.00	0.46	-0.03
potass	-0.07	0.58	0.20	-0.04	0.91	-0.37	0.00	1.00	0.42	-0.50
weight	0.70	0.23	0.22	0.31	0.25	0.14	0.46	0.42	1.00	-0.20
cups	0.09	-0.24	-0.16	0.12	-0.51	0.36	-0.03	-0.50	-0.20	1.00

Appendix:

```
cal<-lm(cereal$rating~cereal$calories)
summary(cal)
Call:
lm(formula = cereal$rating ~ cereal$calories)

Residuals:
     Min       1Q   Median       3Q      Max 
-18.7201  -7.9317  -0.6678   5.9902  23.4161 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)     95.78802    6.55057  14.623  < 2e-16 ***
cereal$calories -0.49701    0.06031  -8.241 4.14e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.24 on 75 degrees of freedom
Multiple R-squared:  0.4752,	Adjusted R-squared:  0.4682 
F-statistic: 67.92 on 1 and 75 DF,  p-value: 4.14e-12

fib<-lm(cereal$rating~cereal$fiber)
summary(fib)
Call:
lm(formula = cereal$rating ~ cereal$fiber)
```

```
Residuals:
    Min      1Q  Median      3Q     Max 
-20.436  -8.159  -2.037   6.491  27.216 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)   35.2566     1.7674  19.948  < 2e-16 ***
cereal$fiber   3.4430     0.5524   6.233 2.45e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.48 on 75 degrees of freedom
Multiple R-squared:  0.3412,	Adjusted R-squared:  0.3325 
F-statistic: 38.85 on 1 and 75 DF,  p-value: 2.445e-08

sug<-lm(cereal$rating~cereal$sugars)
summary(sug)
Call:
lm(formula = cereal$rating ~ cereal$sugars)

Residuals:
    Min      1Q  Median      3Q     Max 
-17.853  -5.677  -1.439   5.160  34.421 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)    59.2844     1.9485   30.43  < 2e-16 ***
cereal$sugars  -2.4008     0.2373  -10.12 1.15e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.196 on 75 degrees of freedom
Multiple R-squared:  0.5771,	Adjusted R-squared:  0.5715 
F-statistic: 102.3 on 1 and 75 DF,  p-value: 1.153e-15



plot(sug$fitted.values, sug$res, xlab="Predicted Rating", ylab="Residual")

qqnorm(sug$residuals)
```

Cooks Distance:

```
cooksd<-cooks.distance(sug)
qf(0.5,2,(77+2))
which(cooksd>  0.6992646)
```

MLR:

```
x1<-(cereal$type == "C")*1
x10<-(cereal$vitamins==25)*1
x11<-(cereal$vitamins==100)*1
x12<-(cereal$shelf==1)*1
x13<-(cereal$shelf==2)*1
x16<-(cereal$mfr_binary=="Popular Brand")*1

full<-
lm(cereal$rating~x1+cereal$calories+cereal$protein+cereal$fat+cereal$sodium+c
ereal$fiber+cereal$carbo+cereal$sugars+cereal$potass+x10+x11+x12+x13+cereal$w
eight+cereal$cups+x16)

null<-lm(cereal$rating~1)
```

Clean cereal dataset for missing values:

```
nutrient_cols <-
c("calories","protein","fat","sodium","fiber","carbo","sugars"
,"potass","vitamins")
cereal[nutrient_cols] <- lapply(cereal[nutrient_cols],function(x) ifelse(x ==
-1, NA, x))
cereal_clean <- na.omit(cereal)
cereal <- cereal_clean
```

Correlations of numerical variables with each other:

```
vars <- c("calories","protein","fat","sodium","fiber","carbo","sugars",
"potass","weight","cups")
nutrients <- cereal[ , vars]
cor_matrix <- cor(nutrients)
round(cor_matrix, 2)
  calories protein   fat sodium fiber carbo sugars potass weight  cups
calories     1.00    0.03  0.51   0.30 -0.30  0.27   0.57  -0.07   0.70  0.09
protein      0.03    1.00  0.20   0.01  0.51 -0.04  -0.29   0.58   0.23 -0.24
fat          0.51    0.20  1.00   0.00  0.01 -0.28   0.29   0.20   0.22 -0.16
sodium       0.30    0.01  0.00   1.00 -0.07  0.33   0.04  -0.04   0.31  0.12
fiber       -0.30    0.51  0.01  -0.07  1.00 -0.38  -0.15   0.91   0.25 -0.51
carbo        0.27   -0.04 -0.28   0.33 -0.38  1.00  -0.45  -0.37   0.14  0.36
sugars       0.57   -0.29  0.29   0.04 -0.15 -0.45   1.00   0.00   0.46 -0.03
potass      -0.07    0.58  0.20  -0.04  0.91 -0.37   0.00   1.00   0.42 -0.50
weight       0.70    0.23  0.22   0.31  0.25  0.14   0.46   0.42   1.00 -0.20
cups         0.09   -0.24 -0.16   0.12 -0.51  0.36  -0.03  -0.50  -0.20  1.00
```

New full model: `full<-lm(cereal$rating~x1+cereal$calories+cereal$protein+cereal$fat+cereal$sodium+cereal$fiber+cereal$carbo+cereal$sugars+x10+x11+x12+x13+cereal$cups+x16)`

```
step(null,scope=list(lower=null, upper=full), direction="both")

final<-lm( cereal$rating ~ cereal$sugars + cereal$fiber + cereal$sodium +
cereal$fat + cereal$protein + x11 + x1 + cereal$carbo + cereal$calories + x13
+ x12)
```

Residual graphs:

```
par(mfrow=c(2,2))


 plot(x1, final$res, pch=19)
 plot(cereal$calories, final$res, pch=19)
 plot(cereal$protein, final$res, pch=19)
 plot(cereal$fat, final$res, pch=19)


 plot(cereal$sodium, final$res, pch=19)
 plot(cereal$fiber, final$res, pch=19)
 plot(cereal$carbo, final$res, pch=19)
 plot(cereal$sugars, final$res, pch=19)

 plot(x11, final$res, pch=19)
 plot(x12, final$res, pch=19)
 plot(x13, final$res, pch=19)
 plot(final$fitted.values, final$res, pch=19)





qqnorm(final$residuals)
```

Standardized Betas:

```
library(lm.beta)
lm.beta(final)
Call:
lm(formula = cereal$rating ~ cereal$sugars + cereal$fiber + cereal$sodium +
    cereal$fat + cereal$protein + x11 + x1 + cereal$carbo + cereal$calories +
    x13 + x12)

Standardized Coefficients::
    (Intercept)   cereal$sugars    cereal$fiber   cereal$sodium
cereal$fat
```

```
             NA     -0.31058741      0.43829086     -0.33847070     -
0.15563346
 cereal$protein             x11              x1    cereal$carbo
cereal$calories
     0.22135524     -0.07036462      0.02393142      0.25008772     -
0.27405674
            x13             x12
     0.02064770      0.01688433
```

Confidence interval:

```
confint(final)
2.5 %      97.5 %
(Intercept)     50.68402854 54.86339109
cereal$sugars   -1.14447929 -0.85532943
cereal$fiber     2.43702243  2.63920948
cereal$sodium   -0.05967102 -0.05510518
cereal$fat      -2.51972270 -1.81889240
cereal$protein   2.64299451  3.13211445
x11             -4.26840799 -2.91787152
x1               1.30288056  4.47524390
cereal$carbo     0.74732925  1.05634632
cereal$calories -0.22769443 -0.15993450
x13              0.13784246  1.15826111
x12              0.06658871  1.01088201
```

VIF:

```
vif(final)
  cereal$sugars    cereal$fiber   cereal$sodium      cereal$fat
cereal$protein
      14.772467        2.232363        1.327996        4.629624
2.574579
            x11              x1    cereal$carbo cereal$calories
x13
       1.280942        1.264577       13.447734       16.811663
1.935696
            x12
       1.603935
```